

# Number Power

Every graded set, every week — 4 weeks, 20 two-page sheets, with an answer key after each week.

WEEK 1

5 sets

WEEK 2

5 sets

WEEK 3

5 sets

WEEK 4

5 sets

## HOW TO USE IT

One page per day. Warm-Up is recall, Core is the graded tier, Challenge is never graded — the score box counts the graded items only. The same items appear in the app's Practice Bay, so a day can be done on paper or on screen, not both.

# How Big Is a Million

Read the place, then say what the digit is worth. Type digits only — no commas needed.

## Warm-Up 11 ITEMS • RECALL

1 How many tens in 100?

\_\_\_\_\_

2 How many hundreds in 1,000?

\_\_\_\_\_

3  $10 \times 100$

\_\_\_\_\_

4  $100 \times 100$

\_\_\_\_\_

5  $1,000 \times 10$

\_\_\_\_\_

6 How many tens in 50

\_\_\_\_\_

7 How many hundreds in 700

\_\_\_\_\_

8  $10 \times 10$

\_\_\_\_\_

9  $100 \times 10$

\_\_\_\_\_

10 Zeros in ten thousand

\_\_\_\_\_

11 How many tens in 1,000

\_\_\_\_\_

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

12. Write six million forty thousand five (6,040,005 — hold the empty places with zeros)

.....

13. The value of the 5 in 5,400

.....

14. The value of the 8 in 380

.....



## Challenge

NOT GRADED • REASON IT OUT

15. How many thousands in 10,000?

16. In 3,472,861 — what is the 4 worth? (It sits in the hundred-thousands place)

17. In 3,472,861 — what is the 7 worth? (Ten-thousands place)

18. In 5,208,043 — what is the 2 worth? (Count the places from the right)

19. How many thousands make 4,000,000? ( $4,000 \times 1,000 = 4,000,000$ )

20. How many zeros in one million? (1,000,000)

21.  $1,000 \times 1,000$  (A thousand thousands)

22. The value of the 2 in 24,000

23. Thousands in 250,000

24. Hundreds in 45,000

25. Millions in one billion

26. Zeros in a hundred million



## Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE



## Error journal

ONE SENTENCE PER MISTAKE

# Ten Times Bigger

Multiplying by ten shifts every digit one place left. Watch the zeros.

## ✦ Warm-Up 9 ITEMS • RECALL

1  $40 \times 10$   
\_\_\_\_\_

2  $700 \times 10$   
\_\_\_\_\_

3  $60 \times 100$   
\_\_\_\_\_

4  $8 \times 1,000$   
\_\_\_\_\_

5  $3,200 \div 10$   
\_\_\_\_\_

6  $20 \times 10$   
\_\_\_\_\_

7  $500 \times 10$   
\_\_\_\_\_

8  $90 \times 100$   
\_\_\_\_\_

9  $4,000 \div 10$   
\_\_\_\_\_

## ◆ Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

10.  $230 \times 100$  (Two more zeros) .....

11.  $9,000 \div 100$  (Take two zeros away) .....

12.  $350 \times 100$  .....

13.  $1,200 \times 10$  .....

14.  $5,600 \div 10$  .....



# Compare, Order, Round

Rounding depends on which place you were asked for. Read the question twice.

## Warm-Up 12 ITEMS • RECALL

1 Round 47 to the nearest ten

\_\_\_\_\_

2 Round 83 to the nearest ten

\_\_\_\_\_

3 Round 65 to the nearest ten

\_\_\_\_\_

4 Round 128 to the nearest ten

\_\_\_\_\_

5 Round 254 to the nearest ten

\_\_\_\_\_

6 Round 999 to the nearest ten

\_\_\_\_\_

7 Round 34 to the nearest ten

\_\_\_\_\_

8 Round 76 to the nearest ten

\_\_\_\_\_

9 Round 55 to the nearest ten

\_\_\_\_\_

10 Round 245 to the nearest hundred

\_\_\_\_\_

11 Round 380 to the nearest hundred

\_\_\_\_\_

12 Round 909 to the nearest ten

\_\_\_\_\_

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. Round 2,499 to the nearest hundred (Look at the tens digit only)

\_\_\_\_\_

14. Round 3,472 to the nearest hundred

\_\_\_\_\_

15. Round 6,509 to the nearest thousand

\_\_\_\_\_

**Challenge** NOT GRADED • REASON IT OUT

16. Round 47,382 to the nearest thousand (382 down beats 618 up)
17. Round 47,382 to the nearest ten thousand (Now the ends are 40,000 and 50,000)
18. Round 863,209 to the nearest hundred thousand (63,209 up to 900,000 is the shorter trip)
19. Larger number: 408,916 or 480,169? Type it. (Compare the ten-thousands place)
20. Largest number that rounds to 5,000 at the nearest thousand (One less than the halfway point)
21. Smallest number that rounds to 5,000 at the nearest thousand (Exactly halfway rounds up)
22. Round 24,650 to the nearest thousand
23. Round 89,999 to the nearest ten thousand
24. Round 12,345 to the nearest hundred
25. Largest whole number that rounds to 700 at the nearest hundred
26. Smallest whole number that rounds to 700 at the nearest hundred

 **Working space** SHOW THE ROOMS, THE GRID, THE NUMBER LINE

**Error journal** ONE SENTENCE PER MISTAKE

# Order of Operations

Build the groups first, then add and subtract what's left loose.

## Warm-Up 12 ITEMS • RECALL

1  $3 + 4 \times 2$   
\_\_\_\_\_

2  $10 - 2 \times 3$   
\_\_\_\_\_

3  $12 \div 4 + 5$   
\_\_\_\_\_

4  $6 + 6 \div 2$   
\_\_\_\_\_

5  $2 \times 3 + 4 \times 5$   
\_\_\_\_\_

6  $20 - (4 + 6)$   
\_\_\_\_\_

7  $5 + 2 \times 3$   
\_\_\_\_\_

8  $8 - 3 \times 2$   
\_\_\_\_\_

9  $16 \div 4 + 1$   
\_\_\_\_\_

10  $9 + 9 \div 3$   
\_\_\_\_\_

11  $(5 + 5) \times 2$   
\_\_\_\_\_

12  $30 - (5 + 5)$   
\_\_\_\_\_

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13.  $5 + 3 \times 4$  (Multiply first:  $3 \times 4 = 12$ )  
.....

14.  $(5 + 3) \times 4$  (Brackets overrule everything)  
.....

15.  $18 \div (3 + 6)$  ( $3 + 6 = 9$  first)  
.....

16.  $40 - 6 \times 5 + 2$  ( $40 - 30 + 2$ )  
.....

17.  $3 \times (12 - 4) \div 6$  ( $3 \times 8 = 24$ , then  $\div 6$ )  
.....

18.  $4 \times 5 - 6 \div 2$   
.....





SHOW THE ROOMS, THE GRID, THE NUMBER LINE



ONE SENTENCE PER MISTAKE

# Powers & Expression Duel

Enrichment day. The little raised number counts copies, not multiples.

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## Challenge

1.  $2^3 (2 \times 2 \times 2)$

2.  $5^2 (5 \times 5)$

3.  $10^4$  (A 1 with four zeros)

4.  $3^2 + 4^2$  (9 + 16)

5.  $2^5 - 2^3$  (32 - 8)

6. Bigger value:  $4^3$  or  $3^4$ ? Type it. (64 against 81)

7.  $10^6 \div 10^3$  (Six zeros take away three)

8. How many zeros in  $10^8$ ? (The exponent counts them)

9.  $3^3$

10.  $4^2$

11.  $5^3$

12.  $2^5$

13.  $6^2$

14.  $2^6$



## Working space



## Error journal

# Week 1 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

## 1.1 • How Big Is a Million

OUT OF 14

Warm-Up • 1-11 • 10 | 10 | 1000 | 10000 | 10000 | 5 | 7 | 100 | 1000 | 4 | 100

Core • 12-14 • 6040005 | 5000 | 80

Challenge • 15-26 • 10 | 400000 | 70000 | 200000 | 4000 | 6 | 1000000 | 20000 | 250 | 450 | 1000 | 8

## 1.2 • Ten Times Bigger

OUT OF 14

Warm-Up • 1-9 • 400 | 7000 | 6000 | 8000 | 320 | 200 | 5000 | 9000 | 400

Core • 10-14 • 23000 | 90 | 35000 | 12000 | 560

Challenge • 15-26 • 45 | 56000 | 74 | 1000 | 100 | 100 | 700 | 60 | 840 | 25000 | 100 | 10000

## 1.3 • Compare, Order, Round

OUT OF 15

Warm-Up • 1-12 • 50 | 80 | 70 | 130 | 250 | 1000 | 30 | 80 | 60 | 200 | 400 | 910

Core • 13-15 • 2500 | 3500 | 7000

Challenge • 16-26 • 47000 | 50000 | 900000 | 480169 | 5499 | 4500 | 25000 | 90000 | 12300 | 749 | 650

## 1.4 • Order of Operations

OUT OF 18

Warm-Up • 1-12 • 11 | 4 | 8 | 9 | 26 | 10 | 11 | 2 | 5 | 12 | 20 | 20

Core • 13-18 • 17 | 32 | 2 | 12 | 4 | 17

Challenge • 22-26 • 11 | 36 | 40 | 50 | 14

## Fri • Powers & Expression Duel

ENRICHMENT

Challenge • 1-14 • 8 | 25 | 10000 | 25 | 24 | 81 | 1000 | 8 | 27 | 16 | 125 | 32 | 36 | 64

# Rounding to Any Place

Find the place, look one to its right, decide.

## Warm-Up 12 ITEMS • RECALL

1 Round 48 to the nearest ten

\_\_\_\_\_

2 Round 43 to the nearest ten

\_\_\_\_\_

3 Round 250 to the nearest hundred

\_\_\_\_\_

4 Round 149 to the nearest hundred

\_\_\_\_\_

5 Round 75 to the nearest ten

\_\_\_\_\_

6 Round 612 to the nearest ten

\_\_\_\_\_

7 Round 62 to the nearest ten

\_\_\_\_\_

8 Round 67 to the nearest ten

\_\_\_\_\_

9 Round 340 to the nearest hundred

\_\_\_\_\_

10 Round 360 to the nearest hundred

\_\_\_\_\_

11 Round 95 to the nearest ten

\_\_\_\_\_

12 Round 450 to the nearest hundred

\_\_\_\_\_

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. Round 4,829 to the nearest hundred

\_\_\_\_\_

14. Round 4,829 to the nearest thousand

\_\_\_\_\_

15. Round 8,950 to the nearest hundred

\_\_\_\_\_

16. Round 5,432 to the nearest thousand

\_\_\_\_\_

17. Round 5,632 to the nearest thousand

\_\_\_\_\_

18. Round 7,777 to the nearest hundred

\_\_\_\_\_



# Estimate Before You Compute

Round first. Then the real answer has something to be checked against.

## Warm-Up 11 ITEMS • RECALL

1 Estimate  $48 + 31$  by rounding both to tens

\_\_\_\_\_

2 Estimate  $62 - 19$

\_\_\_\_\_

3 Estimate  $19 \times 21$

\_\_\_\_\_

4 Estimate  $81 \div 9$

\_\_\_\_\_

5 Estimate  $297 + 104$

\_\_\_\_\_

6 Estimate  $52 + 48$

\_\_\_\_\_

7 Estimate  $39 + 42$  by rounding to tens

\_\_\_\_\_

8 Estimate  $71 - 28$

\_\_\_\_\_

9 Estimate  $29 \times 11$

\_\_\_\_\_

10 Estimate  $63 \div 8$

\_\_\_\_\_

11 Estimate  $198 + 205$

\_\_\_\_\_

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

12. Estimate  $4,812 + 3,190$  to the nearest thousand

.....

13. Estimate  $612 - 388$  to the nearest hundred

.....

14. Estimate  $3,964 \div 4$

.....

15. Estimate  $486 + 317$  to the nearest hundred

.....

16. Estimate  $4,120 - 1,890$  to the nearest hundred

.....

17. Estimate  $396 \times 21$

.....





# How Wrong Is the Estimate

An estimate you never compare to anything teaches you nothing.

## Warm-Up 10 ITEMS • RECALL

1 Estimate of  $48 + 31$ 

\_\_\_\_\_

2 True value of  $48 + 31$ 

\_\_\_\_\_

3 The difference

\_\_\_\_\_

4 True value of  $19 \times 21$ 

\_\_\_\_\_

5 Estimate of  $19 \times 21$ 

\_\_\_\_\_

6 Estimate of  $39 + 42$ 

\_\_\_\_\_

7 True value of  $39 + 42$ 

\_\_\_\_\_

8 True value of  $29 \times 11$ 

\_\_\_\_\_

9 Estimate of  $29 \times 11$ 

\_\_\_\_\_

10 The gap between them

\_\_\_\_\_

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

11. True value of  $612 - 388$ 

.....

12. Estimate of  $612 - 388$  to the nearest hundred

.....

13. How far off was that estimate

.....

14. True value of  $4,812 + 3,190$ 

.....

15. How far off was the thousand estimate

.....

16. Estimate of  $486 + 317$ 

.....

 **Challenge** NOT GRADED • REASON IT OUT

21. Rounding both numbers up always makes the estimate too high or too low — type high or low

22. Estimate  $39 \times 41$ , then give the true value

23. Rounding both numbers down makes the estimate — type high or low

24. Estimate  $49 \times 51$ , then give the true value

 **Working space** SHOW THE ROOMS, THE GRID, THE NUMBER LINE

**Error journal** ONE SENTENCE PER MISTAKE

# Big Numbers in Context

Millions stop being abstract when you attach them to something real.

## Warm-Up 11 ITEMS • RECALL

|                                     |                                  |                                        |                                       |
|-------------------------------------|----------------------------------|----------------------------------------|---------------------------------------|
| 1 Zeros in one thousand<br>_____    | 2 Zeros in one million<br>_____  | 3 Thousands in a million<br>_____      | 4 Hundreds in a thousand<br>_____     |
| 5 Tens in a thousand<br>_____       | 6 Zeros in ten thousand<br>_____ | 7 Zeros in a hundred thousand<br>_____ | 8 Ten thousands in a million<br>_____ |
| 9 Hundreds in ten thousand<br>_____ | 10 Tens in ten thousand<br>_____ | 11 Thousands in ten thousand<br>_____  |                                       |

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

|                                                                                                           |       |
|-----------------------------------------------------------------------------------------------------------|-------|
| 12. Days in about a million minutes, rounded to the nearest hundred (A million minutes is about 694 days) | ..... |
| 13. Dots per page 1,000 — pages for a million dots                                                        | ..... |
| 14. Seconds in an hour                                                                                    | ..... |
| 15. Hours in a week                                                                                       | ..... |
| 16. Minutes in a day                                                                                      | ..... |
| 17. Days in ten years, ignoring leap days                                                                 | ..... |



# A Million Dots, Measured

Prove the estimate. Count one square, scale it up, defend the number.

## ✦ Warm-Up 4 ITEMS • RECALL

1 Dots in a 10 by 10 square

2 Dots in ten such squares

3 Dots in a 5 by 5 square

4 Dots in four such squares

## ◆ Core GRADED • SHOW YOUR WORKING

**STOP RULE** Short Core today — do all of them, then turn the page over.

5. Squares of 100 dots needed for a million

.....

6. A 100 by 100 grid holds how many dots

.....

7. Dots in a 20 by 20 square

.....

8. How many 100 by 100 grids make a million

.....

## ★ Challenge NOT GRADED • REASON IT OUT

9. If one grid is 20 cm wide, the width in metres of 100 side by side

10. Dots in a 1,000 by 1,000 grid

11. Squares of 100 dots to reach 10,000

12. A page of 50 by 40 dots — pages to reach a million

13. Dots on 25 pages of 2,000

14. A 100 by 100 grid — grids needed for a million

## ✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE

## ! Error journal ONE SENTENCE PER MISTAKE

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# Week 2 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

## 2.1 • Rounding to Any Place

OUT OF 18

Warm-Up • 1-12 • 50 | 40 | 300 | 100 | 80 | 610 | 60 | 70 | 300 | 400 | 100 | 500

Core • 13-18 • 4800 | 5000 | 9000 | 5000 | 6000 | 7800

Challenge • 19-26 • 28000 | 100000 | 250 | 349 | 19000 | 46000 | 7500 | 8499

## 2.2 • Estimate Before You Compute

OUT OF 17

Warm-Up • 1-11 • 80 | 40 | 400 | 9 | 400 | 100 | 80 | 40 | 300 | 8 | 400

Core • 12-17 • 8000 | 200 | 1000 | 800 | 2200 | 8000

Challenge • 20-26 • 800 | 748 | 100 | 3072 | 500 | 240 | 3256

## 2.3 • How Wrong Is the Estimate

OUT OF 16

Warm-Up • 1-10 • 80 | 79 | 1 | 399 | 400 | 80 | 81 | 319 | 300 | 19

Core • 11-16 • 224 | 200 | 24 | 8002 | 2 | 800

Challenge • 21-24 • high | 1599 | low | 2499

## 2.4 • Big Numbers in Context

OUT OF 17

Warm-Up • 1-11 • 3 | 6 | 1000 | 10 | 100 | 4 | 5 | 100 | 100 | 1000 | 10

Core • 12-17 • 700 | 1000 | 3600 | 168 | 1440 | 3650

Challenge • 19-22 • 3000 | 100 | 99 | 12

## Fri • A Million Dots, Measured

OUT OF 8

Warm-Up • 1-4 • 100 | 1000 | 25 | 100

Core • 5-8 • 10000 | 10000 | 400 | 100

Challenge • 9-14 • 100 | 20 | 1000000 | 500 | 50000 | 100

# Why Order Matters

Without an agreed order, one expression has several answers.

## ✦ Warm-Up 12 ITEMS • RECALL

1  $2 + 3 \times 4$   
\_\_\_\_\_

2  $(2 + 3) \times 4$   
\_\_\_\_\_

3  $10 - 2 \times 3$   
\_\_\_\_\_

4  $(10 - 2) \times 3$   
\_\_\_\_\_

5  $6 + 6 \div 2$   
\_\_\_\_\_

6  $(6 + 6) \div 2$   
\_\_\_\_\_

7  $3 + 2 \times 5$   
\_\_\_\_\_

8  $(3 + 2) \times 5$   
\_\_\_\_\_

9  $12 - 3 \times 2$   
\_\_\_\_\_

10  $(12 - 3) \times 2$   
\_\_\_\_\_

11  $8 + 8 \div 4$   
\_\_\_\_\_

12  $(8 + 8) \div 4$   
\_\_\_\_\_

## ◆ Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13.  $5 + 4 \times 3 - 2$  .....  
\_\_\_\_\_

14.  $(5 + 4) \times (3 - 2)$  .....  
\_\_\_\_\_

15.  $20 - 3 \times 4 + 2$  .....  
\_\_\_\_\_

16.  $36 \div (2 + 4)$  .....  
\_\_\_\_\_

17.  $36 \div 2 + 4$  .....  
\_\_\_\_\_

18.  $20 - 3 \times 4 + 1$  .....  
\_\_\_\_\_





# Brackets First

Whatever is inside the brackets happens before anything else.

## ✦ Warm-Up 12 ITEMS • RECALL

1  $(4 + 5) \times 2$   
\_\_\_\_\_

2  $(9 - 3) \div 2$   
\_\_\_\_\_

3  $3 \times (2 + 2)$   
\_\_\_\_\_

4  $(7 + 3) \times 10$   
\_\_\_\_\_

5  $(12 - 4) \div 4$   
\_\_\_\_\_

6  $5 \times (10 - 6)$   
\_\_\_\_\_

7  $(2 + 6) \times 3$   
\_\_\_\_\_

8  $(15 - 5) \div 5$   
\_\_\_\_\_

9  $4 \times (3 + 3)$   
\_\_\_\_\_

10  $(9 + 1) \times 10$   
\_\_\_\_\_

11  $(20 - 8) \div 4$   
\_\_\_\_\_

12  $6 \times (12 - 8)$   
\_\_\_\_\_

## ◆ Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13.  $(15 + 5) \div (2 + 2)$  .....

14.  $4 \times (6 + 3) - 10$  .....

15.  $(8 \times 3) - (4 \times 2)$  .....

16.  $2 \times (50 - 15)$  .....

17.  $(25 - 9) \div (2 \times 4)$  .....

18.  $((3 + 3) \times 4) - 10$  .....

## ★ Challenge NOT GRADED • REASON IT OUT

22.  $100 - (25 \times 3)$

23.  $((4 + 2) \times 3) - 8$

24.  $120 \div (2 \times (3 + 3))$

25.  $((10 + 2) \times 5) \div 3$

26.  $240 \div (4 \times (2 + 1))$

 **Working space** SHOW THE ROOMS, THE GRID, THE NUMBER LINE

**Error journal** ONE SENTENCE PER MISTAKE

# Multi-Step Expressions

Multiply and divide before you add and subtract. Left to right within each.

## Warm-Up 12 ITEMS • RECALL

1  $12 \div 3 + 2$

2  $12 + 3 \times 2$

3  $20 \div 4 \times 2$

4  $18 - 6 \div 3$

5  $7 \times 2 + 6$

6  $30 \div 5 - 2$

7  $18 \div 3 + 4$

8  $18 + 3 \times 4$

9  $24 \div 6 \times 2$

10  $20 - 8 \div 4$

11  $9 \times 3 + 3$

12  $40 \div 8 - 3$

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13.  $48 \div 6 + 3 \times 4$

14.  $7 \times 8 - 6 \times 5$

15.  $64 \div 8 \div 2$

16.  $9 + 36 \div 4 - 3$

17.  $60 \div 5 + 7 \times 2$

18.  $81 \div 9 - 3 \times 2$



# Write the Expression

Turn the sentence into symbols, then work it out.

## Warm-Up 12 ITEMS • RECALL

1 Five more than three lots of four

\_\_\_\_\_

2 Double seven, then add one

\_\_\_\_\_

3 Ten less than six times three

\_\_\_\_\_

4 Half of twenty, plus four

\_\_\_\_\_

5 Three lots of nine

\_\_\_\_\_

6 Twenty shared by four

\_\_\_\_\_

7 Four more than double six

\_\_\_\_\_

8 Triple five, then add two

\_\_\_\_\_

9 Twenty less than five times six

\_\_\_\_\_

10 A third of thirty, plus one

\_\_\_\_\_

11 Two lots of nine, minus eight

\_\_\_\_\_

12 Half of sixteen, times three

\_\_\_\_\_

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. The sum of 8 and 4, multiplied by 3

.....

14. 8 added to 4 lots of 3

.....

15. 100 less the product of 7 and 9

.....

16. The difference of 20 and 8, divided by 4

.....

17. Six lots of the sum of 2 and 3

.....

18. Six times the sum of two and five

.....



## Working space



## Error journal

# Mid-Unit Quiz

Eight items across Weeks 1–3. 85% to keep flying.

## ✦ Warm-Up 4 ITEMS • RECALL

1 Round 4,829 to the nearest hundred  
\_\_\_\_\_2  $2 + 3 \times 4$   
\_\_\_\_\_3 Round 3,451 to the nearest hundred  
\_\_\_\_\_4  $5 + 2 \times 3$   
\_\_\_\_\_

## ◆ Core GRADED • SHOW YOUR WORKING

**STOP RULE** This one is a test. Do **every** problem — no stopping early.5.  $(5 + 4) \times (3 - 2)$  .....6.  $48 \div 6 + 3 \times 4$  .....

7. Zeros in one million .....

8. The sum of 8 and 4, multiplied by 3 .....

9.  $(20 - 3) \times 4$  .....10. Estimate  $396 \times 21$  .....11. Estimate  $187 \times 4$  .....





## Working space



## Error journal

# Week 3 Answers

Mark the reasoning, not the handwriting.

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

## 3.1 • Why Order Matters

OUT OF 18

Warm-Up • 1-12 • 14 | 20 | 4 | 24 | 9 | 6 | 13 | 25 | 6 | 18 | 10 | 4

Core • 13-18 • 15 | 9 | 10 | 6 | 22 | 9

Challenge • 23-26 • 70 | 50 | 54 | 30

## 3.2 • Brackets First

OUT OF 18

Warm-Up • 1-12 • 18 | 3 | 12 | 100 | 2 | 20 | 24 | 2 | 24 | 100 | 3 | 24

Core • 13-18 • 5 | 26 | 16 | 70 | 2 | 14

Challenge • 22-26 • 25 | 10 | 10 | 20 | 20

## 3.3 • Multi-Step Expressions

OUT OF 18

Warm-Up • 1-12 • 6 | 18 | 10 | 16 | 20 | 4 | 10 | 30 | 8 | 18 | 30 | 2

Core • 13-18 • 20 | 26 | 4 | 15 | 26 | 3

Challenge • 20-26 • 52 | 31 | 68 | 33 | 36 | 58 | 98

## 3.4 • Write the Expression

OUT OF 18

Warm-Up • 1-12 • 17 | 15 | 8 | 14 | 27 | 5 | 16 | 17 | 10 | 11 | 10 | 24

Core • 13-18 • 36 | 20 | 37 | 3 | 30 | 42

Challenge • 23-26 • 24 | 4 | 48 | 10

## Fri • Mid-Unit Quiz

OUT OF 11

Warm-Up • 1-4 • 4800 | 14 | 3500 | 11

Core • 5-11 • 9 | 20 | 800 | 6 | 36 | 68 | 8000

Challenge • 12-15 • 31 | 14 | 749 | 86400

# The Little Raised Number

An exponent counts how many times the base is multiplied by itself.

## ✦ Warm-Up 10 ITEMS • RECALL

1

 $2^2$ 

\_\_\_\_\_

2

 $3^2$ 

\_\_\_\_\_

3

 $5^2$ 

\_\_\_\_\_

4

 $2^3$ 

\_\_\_\_\_

5

 $4^2$ 

\_\_\_\_\_

6

 $6^2$ 

\_\_\_\_\_

7

 $7^2$ 

\_\_\_\_\_

8

 $8^2$ 

\_\_\_\_\_

9

 $9^2$ 

\_\_\_\_\_

10

 $1^5$ 

\_\_\_\_\_

## ◆ Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

11.  $3^3$ 

.....

12.  $2^4$ 

.....

13.  $5^3$ 

.....

14.  $2^5$ 

.....

15.  $4^3$ 

.....

16.  $11^2$ 

.....

 **Challenge** NOT GRADED • REASON IT OUT

17.  $10^2$

18.  $10^3$

19.  $2^8$

20. Which is bigger,  $3^4$  or  $4^3$  — type the value of the bigger

21.  $2^{10}$

22. Which is bigger,  $2^6$  or  $6^2$  — type the value of the bigger

 **Working space** SHOW THE ROOMS, THE GRID, THE NUMBER LINE

**Error journal** ONE SENTENCE PER MISTAKE

# Powers of Ten

The exponent is the number of zeros. That is the whole shortcut.

## Warm-Up 3 ITEMS • RECALL

1  $10^1$ 2 Zeros in  $10^7$ 3  $10^0$ 

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

4. Write 5,000 as a digit times a power of ten — type the exponent

.....



## Challenge NOT GRADED • REASON IT OUT

NOT GRADED • REASON IT OUT

5.  $10^2$

6.  $10^3$

$7. \cdot 10^4$

8. Zeros in  $10^5$

9.  $10^6$

10.  $3 \times 10^2$

11.  $7 \times 10^3$

12.  $4 \times 10^4$

13.  $25 \times 10^2$

14.  $10^3 \times 10^2$

15. A million written as a power of ten — type the exponent

16.  $10^5$



## Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE



## Error journal

ONE SENTENCE PER MISTAKE

# Expanded Form with Powers

Every place value is a power of ten in disguise.

## Warm-Up 12 ITEMS • RECALL

1 The value of the 4 in 400

\_\_\_\_\_

2 The value of the 7 in 70

\_\_\_\_\_

3 The value of the 3 in 3,000

\_\_\_\_\_

4  $200 + 30 + 5$

\_\_\_\_\_

5  $4,000 + 200 + 10$

\_\_\_\_\_

6 The value of the 9 in 900

\_\_\_\_\_

7 How many hundreds make 2,600

\_\_\_\_\_

8 How many hundreds make 4,300

\_\_\_\_\_

9 How many hundreds make 7,100

\_\_\_\_\_

10 How many hundreds make 9,500

\_\_\_\_\_

11 How many tens make 3,000

\_\_\_\_\_

12 How many tens make 6,000

\_\_\_\_\_

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

20. In 4,271 — what is the 4 worth

.....

21. In 4,271 — what is the 2 worth

.....

22. In 4,271 — what is the 7 worth

.....

23. In 8,605 — what is the 8 worth

.....

24. In 8,605 — what is the 6 worth

.....

25. In 3,940 — what is the 9 worth

.....



## Challenge

NOT GRADED • REASON IT OUT

33.  $6 \times 10^3 + 4 \times 10^2 + 2$

34.  $3 \times 10^4 + 5 \times 10^2$

35. The value of the 8 in 84,000

36.  $9 \times 10^2 + 9 \times 10^1 + 9$

37.  $7 \times 10^5$

38.  $2 \times 10^5 + 6 \times 10^3 + 4 \times 10^1$

39. The largest power of ten inside 47,300 — type its exponent

40.  $5 \times 10^3 + 2 \times 10^2$

41.  $7 \times 10^4 + 3 \times 10^2$

42.  $4 \times 10^3 + 6 \times 10^1 + 2$

43.  $20,000 + 4,000 + 300 + 50$

44.  $3 \times 10^5 + 7 \times 10^3 + 5 \times 10^1$



## Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE



## Error journal

ONE SENTENCE PER MISTAKE



# Error Journal Sweep

Mixed review of the whole mission. Fix only what repeats.

## Warm-Up 12 ITEMS • RECALL

1 Round 149 to the nearest hundred

\_\_\_\_\_

2  $2 + 3 \times 4$

\_\_\_\_\_

3 Zeros in one million

\_\_\_\_\_

4  $5^2$

\_\_\_\_\_

5  $(2 + 3) \times 4$

\_\_\_\_\_

6 Round 340 to the nearest hundred

\_\_\_\_\_

7  $5 + 2 \times 3$

\_\_\_\_\_

8 Zeros in ten thousand

\_\_\_\_\_

9  $6^2$

\_\_\_\_\_

10  $(3 + 2) \times 5$

\_\_\_\_\_

11  $48 \div 6 + 3 \times 4$

\_\_\_\_\_

12 Estimate  $3,964 \div 4$

\_\_\_\_\_

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. Estimate  $486 + 317$  to the nearest hundred

.....

14.  $60 \div 5 + 7 \times 2$

.....

15.  $2^4$

.....

16.  $10^3$

.....

17. Round 96,412 to the nearest ten thousand

.....

18.  $4 \times 10^4$

.....

19.  $6 \times 10^3 + 4 \times 10^2 + 2$

.....

20.  $144 \div 12 + 8 \times 3 - 5$

.....

21.  $2^8$

.....

22.  $10^4$

.....



# Mission 03 Test

Twelve items plus the Big Question, answered out loud.

## Warm-Up 4 ITEMS • RECALL

1 Round 612 to the nearest ten  
\_\_\_\_\_2  $3^2$   
\_\_\_\_\_3 Round 448 to the nearest ten  
\_\_\_\_\_4  $4^2$   
\_\_\_\_\_

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** This one is a test. Do **every** problem — no stopping early.5. Round 4,829 to the nearest thousand  
.....6.  $10 - 2 \times 3$   
.....7.  $(15 + 5) \div (2 + 2)$   
.....8.  $2^5$   
.....9. Estimate  $4,812 + 3,190$  to the nearest thousand  
.....10. The sum of 12 and 8, divided by the difference of 9 and 4  
.....11.  $(15 - 5) \div 5$   
.....12.  $100 - 4 \times 12$   
.....13.  $7 \times 10^3$   
.....14.  $2 \times 10^5 + 6 \times 10^3 + 4 \times 10^1$   
.....

**Challenge** NOT GRADED • REASON IT OUT

15.  $2 \times 3 + 4 \times 5 + 6 \times 7$

16.  $7 \times 10^4$

17.  $3 \times 4 + 5 \times 6 + 7 \times 8$

18. A million seconds in days, to the nearest whole

19.  $5^3$



## Working space



## Error journal

# Week 4 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

## 4.1 • The Little Raised Number

OUT OF 16

Warm-Up • 1-10 • 4 | 9 | 25 | 8 | 16 | 36 | 49 | 64 | 81 | 1

Core • 11-16 • 27 | 16 | 125 | 32 | 64 | 121

Challenge • 17-22 • 100 | 1000 | 256 | 81 | 1024 | 64

## 4.2 • Powers of Ten

OUT OF 4

Warm-Up • 1-3 • 10 | 7 | 1

Core • 4 • 3

Challenge • 5-16 • 100 | 1000 | 10000 | 5 | 1000000 | 300 | 7000 | 40000 | 2500 |  
100000 | 6 | 100000

## 4.3 • Expanded Form with Powers

OUT OF 18

Warm-Up • 1-12 • 400 | 70 | 3000 | 235 | 4210 | 900 | 26 | 43 | 71 | 95 | 300 |  
600

Core • 20-25 • 4000 | 200 | 70 | 8000 | 600 | 900

Challenge • 33-44 • 6402 | 30500 | 80000 | 999 | 700000 | 206040 | 4 | 5200 | 70300 |  
4062 | 24350 | 307050

## Thu • Error Journal Sweep

OUT OF 22

Warm-Up • 1-12 • 100 | 14 | 1000 | 6 | 25 | 20 | 300 | 11 | 10000 | 4 | 36 | 25

Core • 13-22 • 100000 | 20 | 40000 | 1000 | 6402 | 800 | 26 | 25000 | 4062 | 16

Challenge • 23-26 • 31 | 256 | 58 | 1024

## Fri • Mission 03 Test

OUT OF 14

Warm-Up • 1-4 • 610 | 9 | 450 | 16

Core • 5-14 • 5000 | 4 | 5 | 52 | 7000 | 32 | 8000 | 4 | 2 | 70000

Challenge • 15-19 • 206040 | 68 | 98 | 12 | 125